

Executive Profitability Dashboard – System Analysis & Design

Private Bus Company | Data, Engineering & Architecture Blueprint

1. Business Context & Objectives

- Provide executives with immediate visibility into where money is made or lost.
- Enable strategic decisions on route expansion, cancellation, fleet investment, and marketing.
- Deliver trusted, decision-ready KPIs in under two minutes.

2. Scope & Boundaries

- In scope: executive KPIs, route profitability, cost efficiency, occupancy, fuel analysis.
- Out of scope (Phase 2+): dynamic pricing, real-time GPS, automated mobile money integration.

3. Stakeholders & Users

- CEO / Owners – strategy and profitability oversight.
- COO – operational drilldowns.
- Finance – margin and cost validation.
- Fleet Manager – utilization and fuel efficiency.

4. Core KPIs

- Total Revenue (Daily / Weekly / Monthly).
- Net Profit or Contribution Margin.
- Revenue per Route.
- Cost per Kilometer.
- Average Occupancy Rate (Load Factor).
- Fuel Cost as Percentage of Revenue.

5. Decisions Enabled

- Expand high-margin routes.
- Fix or cancel loss-making routes.
- Reallocate buses to high-demand corridors.
- Invest in marketing or fleet based on ROI.

6. Data Architecture

- Source systems: ticketing, trips, payments, fuel logs, fleet data.
- Warehouse: PostgreSQL / BigQuery.
- Transformation: SQL/dbt curated models.
- Serving layer: BI dashboard or web analytics API.

7. KPI Definitions (Summary)

- Revenue = Sum of successful payments.
- Load Factor = Passengers / Seats offered.
- Cost per KM = Direct costs / Total kilometers.
- Fuel % = Fuel cost / Revenue.
- Margin = (Revenue – Direct costs) / Revenue.

8. Dashboard Layout

- Top KPI tiles with trends and deltas.
- Route profitability ranking table.
- Revenue, margin, fuel and occupancy trends.
- Alerts for loss-making routes and cost anomalies.

9. Data Quality & Governance

- Completeness and validity checks on trips, payments, fuel.
- Clear KPI definitions approved by Finance and CEO.
- Daily refresh with monitoring and alerts.

10. Delivery Phases

- Phase 1: KPI dashboard + curated data model.
- Phase 2: Cost realism and alerting.
- Phase 3: Forecasting and optimization.